

Week 5 Assignment Question

1. Which predefined method in the scikit-learn library is used for multiple linear regression?

- a. LinearRegression()
- b. PolynomialRegression()
- c. MultiRegression()
- d. Regression()

Answer: a LinearRegression()

Explanation: The scikit-learn library implements both simple and multiple linear regression using the same class LinearRegression(), the only difference is the size of input features given.

2. Let N learning examples were given to a regression model. If y_i and $\hat{y}_i$ are the actual and predicted values, respectively, corresponding to the i^{th} input, what will be the correct expression for root mean square error metric?

- a. $\frac{1}{\sqrt{N}} \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2}$
- b. $\frac{1}{\sqrt{N}} \sum_{i=1}^N \sqrt{(y_i - \hat{y}_i)^2}$
- c. $\frac{1}{N} \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2}$
- d. $\sqrt{\frac{1}{\sqrt{N}} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$

Answer: a

Explanation: The correct expression for RMSE is $\frac{1}{\sqrt{N}} \sqrt{\sum_{i=1}^N (y_i - \hat{y}_i)^2}$ which is the square root of the mean difference between the predicted and actual values of the learning model outputs.

3. According to the ordinary least squares method, what will be the correct implementation of the beta value of multiple linear regression model using predefined function of numpy library (as np), given that X is the input value matrix and y is the correct output value vector?

- a. `np.linalg.inv(np.dot(X.T, X_train)).dot(X).dot(y.T)`
- b. `np.linalg.inv(np.dot(X.T, X)).dot(X.T).dot(y)`
- c. `np.linalg.inv(np.dot(X.T, y)).dot(X.T).dot(y)`
- d. `np.linalg.inv(np.dot(X.T, X.T)).dot(X.T).dot(y)`

Answer: b

Explanation: The correct implementation of solution of ordinary least square method is given by expression $\beta = (X^T X)^{-1} X^T y$ which can be correctly implemented in python using the numpy library as `np.linalg.inv(np.dot(X.T, X)).dot(X.T).dot(y)`

4. Consider the following statements in regards to polynomial regression?

- I. If the degree is too low, the model will underfit the data.
- II. If the degree is too low, the model will overfit the data
- III. If the degree is too high, the model will underfit the data.
- IV. If the degree is too high, the model will overfit the data.

Which of these statements is true?

- a. Only statement (I) is true.
- b. Only statement (III) is true.
- c. Statement (I) and (IV) are true.
- d. Statement (II) and (III) are true.

Answer: c.

Explanation: The degree in polynomial regression decides the expressibility of the learning model, higher the degree means the model is more expressive which might lead to overfitting on train data, which in turn affects its performance on unseen data. On the other hand, if the degree is too low, it will affect the model's performance on the training data as the expressibility of the learning model will be very low.

5. If $f(X) = m_0 + m_1x + m_2x^2 + m_3x^3$. What type of regression is this?

- a) Polynomial regression
- b) Simple linear regression
- c) Logistic regression
- d) Multiple linear regression

Answer: a

Explanation: The given function corresponds to the polynomial regression with one independent variable which is scaled by introducing two more terms of quadratic and cubic powers of the independent variables. This helps the regressor to learn non-linear feature spaces and better approximate the inherent data distribution in a given non-linear dataset.

Week 5 Assignment Questions

1. What is the primary purpose of the 'LinearRegression()' class in the scikit-learn library?

- a) To perform only simple linear regression
- b) To perform both simple and multiple linear regression
- c) To perform only polynomial regression
- d) To perform logistic regression

Ans: b) To perform both simple and multiple linear regression

Explanation: The 'LinearRegression()' class in scikit-learn can be used for both simple and multiple linear regression, depending on the number of input features provided.

2. What does Root Mean Square Error (RMSE) measure in regression models?

- a) The sum of absolute differences between actual and predicted values
- b) The variance of predicted values
- c) The square root of the average squared difference between actual and predicted values
- d) The correlation between actual and predicted values

Ans: c) The square root of the average squared difference between actual and predicted values

Explanation: RMSE measures how far the model's predictions are from actual values, providing a useful error metric that penalizes large errors more than smaller ones.

3. What is the role of the beta coefficients (β) in multiple linear regression?

- a) They indicate the strength and direction of the relationship between each independent variable and the dependent variable
- b) They determine the probability of an event occurring in logistic regression
- c) They group similar data points together in a clustering algorithm
- d) They help in reducing the dimensionality of the dataset

Ans: a) They indicate the strength and direction of the relationship between each independent variable and the dependent variable

Explanation: In multiple linear regression, **beta coefficients (β)** represent how much the dependent variable is expected to change when an independent variable changes by one unit, while keeping other variables constant. These coefficients determine the impact of each predictor on the target variable.

4. How does the degree of a polynomial regression model affect its performance?

- a) A lower-degree polynomial leads to overfitting, and a higher-degree polynomial leads to underfitting
- b) A lower-degree polynomial leads to underfitting, and a higher-degree polynomial leads to overfitting
- c) The degree of the polynomial does not affect the performance of the model
- d) Polynomial regression is always better than linear regression

Ans: b) A lower-degree polynomial leads to underfitting, and a higher-degree polynomial leads to overfitting

Explanation: If the polynomial degree is too low, the model will fail to capture the complexity of the data (underfitting). If the degree is too high, the model will fit the training data too well but may not generalize to unseen data (overfitting).

5. What is the main difference between multiple linear regression and polynomial regression?

- a) Multiple linear regression allows for multiple independent variables, whereas polynomial regression is restricted to one independent variable
- b) Multiple linear regression models a linear relationship, while polynomial regression can model nonlinear relationships
- c) Multiple linear regression always performs better than polynomial regression
- d) There is no difference between multiple linear regression and polynomial regression

Ans: b) Multiple linear regression models a linear relationship, while polynomial regression can model nonlinear relationships

Explanation: Multiple linear regression assumes a linear relationship between dependent and independent variables, while polynomial regression introduces higher-degree terms to capture nonlinear patterns.